THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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ANSWER KEY & MARKING SCHEME SET 1

Chapter 11: **Exploring Some Geometric Themes** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 8	$n = 6$, faces $= n + 2$
2	**(d)** 12	Base has $n = 6$ sides; edges $= 2n$
3	**(a)** 12	Each edge is shared by 2 faces: $E = 3F/2$
4	**(d)** 20	$V = 2 + E - F$
5	**(c)** 8	Inner cube of side 2
6	**(d)** 48	Length multiplies by $\frac{4}{3}$ each stage: $27 \times (4/3)^2$
7	**(a)** 13	$1 + 3 + \cdots + 3^2 = (3^3 - 1)/2$
8	**(b)** 10	$n = 8$, faces $= n + 2$
9	**(d)** 16	Base has $n = 8$ sides; edges $= 2n$
10	**(d)** 18	Each edge is shared by 2 faces: $E = 3F/2$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	12 edges (1); sum $= 4(15 + 9 + 6) = 120$ cm (1).	1+1
12	$n = 12$ (1); faces 14, vertices 24 (1).	1+1
13	Segments $= 16$ (1); total $= 112$ cm (1).	1+1
14	12 edges (1); sum $= 4(14 + 8 + 5) = 108$ cm (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	(i) 8 (1); (ii) $(4 - 2)^3 = 8$ (1); (iii) $64 - 8 = 56$ (1).	1+1+1
16	$E = 3F/2 = 15$ (1); $V = 2 + E - F = 7$ (2).	1+2
17	Number $= 3^3 = 27$ (1); shaded area $= 128 \times (3/4)^3 = 54$ cm² (2).	1+2
	OR: (i) 8 (1); (ii) $(5 - 2)^3 = 27$ (1); (iii) $125 - 27 = 98$ (1).	1+1+1
18	$E = 3F/2 = 27$ (1); $V = 2 + E - F = 11$ (2).	1+2
	OR: Number $= 3^2 = 9$ (1); shaded area $= 320 \times (3/4)^2 = 180$ cm² (2).	1+2

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Prism: $7, 10, 15$; pyramid: $6, 6, 10$ (2); differences: faces 1 (1), vertices 4 (1), edges 5 (1).	2+1+1+1
	OR: Stage 0: $3 \times 9 = 27$ cm (1); stage 1: $27 \times \frac{4}{3} = 36$ cm (2); stage 2: 48 cm (2).	1+2+2
20	Prism: $8, 12, 18$; pyramid: $7, 7, 12$ (2); differences: faces 1 (1), vertices 5 (1), edges 6 (1).	2+1+1+1
	OR: Stage 0: $3 \times 18 = 54$ cm (1); stage 1: $54 \times \frac{4}{3} = 72$ cm (2); stage 2: 96 cm (2).	1+2+2

Mistakes to avoid:

- Counting faces, edges and vertices of a prism and pyramid with each other's formulas (a pyramid has $2n$ edges, a prism $3n$).
- Forgetting the apex when counting the vertices of a pyramid (a square pyramid has 5, not 4).
- In painted-cube questions, using $n - 2$ for the edge cubes but forgetting to multiply by the number of edges (12) or faces (6).
- Assuming that any six squares joined edge to edge fold into a cube; the net must not overlap when folded.
- Using Euler's formula as $F + V + E = 2$; the correct form is $F + V - E = 2$.
- Counting the stages of a fractal wrongly: the starting figure is stage 0, not stage 1.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 2

Chapter 11: **Exploring Some Geometric Themes** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 60	$V = 2 + E - F$
2	**(b)** 64	Inner cube of side 4
3	**(a)** 192	Length multiplies by $\frac{4}{3}$ each stage: $81 \times (4/3)^3$
4	**(d)** 40	$1 + 3 + \cdots + 3^3 = (3^4 - 1)/2$
5	**(b)** 11	$n = 9$, faces $= n + 2$
6	**(d)** 20	Base has $n = 10$ sides; edges $= 2n$
7	**(c)** 21	Each edge is shared by 2 faces: $E = 3F/2$
8	**(b)** 24	$V = 2 + E - F$
9	**(c)** 216	Inner cube of side 6
10	**(a)** 448	Length multiplies by $\frac{4}{3}$ each stage: $189 \times (4/3)^3$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$n = 13$ (1); faces 15, vertices 26 (1).	1+1
12	Segments $= 16$ (1); total $= 128$ cm (1).	1+1
13	12 edges (1); sum $= 4(18 + 11 + 7) = 144$ cm (1).	1+1
14	$n = 15$ (1); faces 17, vertices 30 (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	(i) 8 (1); (ii) $(6 - 2)^3 = 64$ (1); (iii) $216 - 64 = 152$ (1).	1+1+1
16	$E = 3F/2 = 33$ (1); $V = 2 + E - F = 13$ (2).	1+2
17	Number $= 3^4 = 81$ (1); shaded area $= 1280 \times (3/4)^4 = 405$ cm² (2).	1+2
	OR: (i) 8 (1); (ii) $(7 - 2)^3 = 125$ (1); (iii) $343 - 125 = 218$ (1).	1+1+1
18	$E = 3F/2 = 39$ (1); $V = 2 + E - F = 15$ (2).	1+2
	OR: Number $= 3^2 = 9$ (1); shaded area $= 576 \times (3/4)^2 = 324$ cm² (2).	1+2

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Prism: $9, 14, 21$; pyramid: $8, 8, 14$ (2); differences: faces 1 (1), vertices 6 (1), edges 7 (1).	2+1+1+1
	OR: Stage 0: $3 \times 27 = 81$ cm (1); stage 1: $81 \times \frac{4}{3} = 108$ cm (2); stage 2: 144 cm (2).	1+2+2
20	Prism: $10, 16, 24$; pyramid: $9, 9, 16$ (2); differences: faces 1 (1), vertices 7 (1), edges 8 (1).	2+1+1+1
	OR: Stage 0: $3 \times 36 = 108$ cm (1); stage 1: $108 \times \frac{4}{3} = 144$ cm (2); stage 2: 192 cm (2).	1+2+2

Mistakes to avoid:

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- Forgetting the apex when counting the vertices of a pyramid (a square pyramid has 5, not 4).
- In painted-cube questions, using $n - 2$ for the edge cubes but forgetting to multiply by the number of edges (12) or faces (6).
- Assuming that any six squares joined edge to edge fold into a cube; the net must not overlap when folded.
- Using Euler's formula as $F + V + E = 2$; the correct form is $F + V - E = 2$.
- Counting the stages of a fractal wrongly: the starting figure is stage 0, not stage 1.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 11: **Exploring Some Geometric Themes** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 121	$1+3+\cdots+3^4=(3^5-1)/2$
2	**(c)** 12	$n=10$, faces $=n+2$
3	**(d)** 24	Base has $n=12$ sides; edges $=2n$
4	**(c)** 24	Each edge is shared by 2 faces: $E=3F/2$
5	**(a)** 12	$V=2+E-F$
6	**(b)** 343	Inner cube of side 7
7	**(d)** 64	Length multiplies by $\frac{4}{3}$ each stage: $36\times(4/3)^2$
8	**(d)** 364	$1+3+\cdots+3^5=(3^6-1)/2$
9	**(c)** 14	$n=12$, faces $=n+2$
10	**(b)** 30	Base has $n=15$ sides; edges $=2n$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Segments $=16$ (1); total $=144$ cm (1).	1+1
12	12 edges (1); sum $=4(20+12+9)=164$ cm (1).	1+1
13	$n=16$ (1); faces 18, vertices 32 (1).	1+1
14	Segments $=16$ (1); total $=160$ cm (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	(i) 8 (1); (ii) $(9-2)^3=343$ (1); (iii) $729-343=386$ (1).	1+1+1
16	$E=3F/2=42$ (1); $V=2+E-F=16$ (2).	1+2
17	Number $=3^5=243$ (1); shaded area $=2048\times(3/4)^5=486$ cm² (2).	1+2
	OR: (i) 8 (1); (ii) $(10-2)^3=512$ (1); (iii) $1000-512=488$ (1).	1+1+1
18	$E=3F/2=45$ (1); $V=2+E-F=17$ (2).	1+2
	OR: Number $=3^3=27$ (1); shaded area $=4096\times(3/4)^3=1728$ cm² (2).	1+2

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Prism: 11, 18, 27; pyramid: 10, 10, 18 (2); differences: faces 1 (1), vertices 8 (1), edges 9 (1).	2+1+1+1
	OR: Stage 0: $3\times45=135$ cm (1); stage 1: $135\times\frac{4}{3}=180$ cm (2); stage 2: 240 cm (2).	1+2+2
20	Prism: 12, 20, 30; pyramid: 11, 11, 20 (2); differences: faces 1 (1), vertices 9 (1), edges 10 (1).	2+1+1+1
	OR: Stage 0: $3\times54=162$ cm (1); stage 1: $162\times\frac{4}{3}=216$ cm (2); stage 2: 288 cm (2).	1+2+2

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- Using Euler's formula as $F+V+E=2$; the correct form is $F+V-E=2$.
- Counting the stages of a fractal wrongly: the starting figure is stage 0, not stage 1.

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